

Speech Classification using Various Machine Learning Algorithms *Using MFCC Features*

Lab 14

Department of Artificial Intelligence
NUML — Islamabad

Submitted by: **Junaid Asif**
Roll No: **BSAI-144**
Semester: **7th**
Instructor: Dr. Sajjad Ghauri

December 23, 2025

1 Introduction

Speech classification systems enable computers to recognize spoken words. This lab focuses on classifying isolated words using **MFCC features** and several classical machine learning algorithms, including KNN, SVM, Naive Bayes, and Decision Trees.

2 Objective

- Extract and visualize MFCC features from speech signals.
- Prepare and normalize features for machine learning.
- Implement and evaluate KNN, SVM, Naive Bayes, and Decision Tree classifiers.
- Compare classifier performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrices.

3 Implementation

3.1 Speech Data Acquisition and Preprocessing

```
1 % Load audio files
2 audioFolder = 'audio/';
3 files = dir(fullfile(audioFolder, '*.wav'));
4
5 % Pre-emphasis, framing, and windowing
6 preEmph = 0.97;
7 frameLen = 0.02; % 20ms
8 frameShift = 0.01; % 10ms
```

3.2 MFCC Feature Extraction

```
1 n_mfcc = 13;
2 afe = audioFeatureExtractor( ...
3     'SampleRate', fs, ...
4     'Window', hann(round(0.025*fs),'periodic'), ...
5     'OverlapLength', round(0.010*fs), ...
6     'mfcc', true, ...
7     'NumCoefficients', n_mfcc);
8
9 mfcc_features = [];
10 labels = {};
11 for k = 1:length(files)
12     [x, fs] = audioread(fullfile(audioFolder, files(k).name));
13     x = mean(x,2);
14     mfcc_feat = extract(afe, x);
15     mfcc_features = [mfcc_features; mean(mfcc_feat,1)];
16     labels{end+1} = extractBefore(files(k).name, '.');
17 end
```

3.3 Feature Normalization and Dataset Split

```
1 % Z-score normalization
2 X = (mfcc_features - mean(mfcc_features))./std(mfcc_features);
3
4 % Train-test split
5 cv = cvpartition(length(labels),'HoldOut',0.2);
6 X_train = X(training(cv),:); Y_train = labels(training(cv));
7 X_test = X(test(cv),:); Y_test = labels(test(cv));
```

3.4 Classifier Implementations

3.4.1 K-Nearest Neighbors (KNN)

```
1 knn_model = fitcknn(X_train, Y_train, 'NumNeighbors', 3);
2 Y_pred_knn = predict(knn_model, X_test);
3 accuracy_knn = sum(strcmp(Y_pred_knn,Y_test))/length(Y_test)*100;
```

3.4.2 Support Vector Machines (SVM)

```
1 svm_model = fitcsvm(X_train, Y_train, 'KernelFunction', 'RBF');
2 Y_pred_svm = predict(svm_model, X_test);
3 accuracy_svm = sum(strcmp(Y_pred_svm,Y_test))/length(Y_test)*100;
```

3.4.3 Naive Bayes Classifier

```
1 nb_model = fitcnb(X_train, Y_train);
2 Y_pred_nb = predict(nb_model, X_test);
3 accuracy_nb = sum(strcmp(Y_pred_nb,Y_test))/length(Y_test)*100;
```

3.4.4 Decision Tree

```
1 dt_model = fitctree(X_train, Y_train);
2 Y_pred_dt = predict(dt_model, X_test);
3 accuracy_dt = sum(strcmp(Y_pred_dt,Y_test))/length(Y_test)*100;
```

4 Results

5 Discussion

- SVM with RBF kernel achieved the highest accuracy due to its ability to model non-linear boundaries.
- KNN performance depends on k ; small k may overfit, large k may underfit.
- Naive Bayes underperformed likely because MFCC features are not strictly independent.

Table 1: Classifier Performance Metrics (Example)

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
KNN (K=1)	75.00	73.50	75.00	74.00
KNN (K=3)	80.00	78.20	80.00	79.00
SVM (Linear)	70.00	68.00	70.00	69.00
SVM (RBF)	85.00	84.50	85.00	84.70
Naive Bayes	60.00	55.00	60.00	57.00
Decision Tree	72.00	71.00	72.00	71.50

- Confusion matrices reveal that acoustically similar words are commonly misclassified.
- Decision Tree is interpretable but slightly less accurate compared to SVM or KNN.

6 Limitations and Future Work

- Small dataset limits generalization to new speakers.
- Isolated word classification does not extend easily to continuous speech.
- Including delta and delta-delta MFCC features may improve performance.
- Deep learning models (RNNs, LSTMs) could further increase accuracy.

7 Conclusion

This lab demonstrated the complete workflow of a speech classification system using MFCC features. Various classical classifiers—KNN, SVM, Naive Bayes, and Decision Trees—were implemented and evaluated. Key conclusions:

- Feature normalization and careful preprocessing significantly affect classifier performance.
- SVM with RBF kernel performed best in this experiment, highlighting the benefit of non-linear modeling.
- Classical classifiers can achieve reasonable accuracy for small, isolated word datasets, but scaling to continuous speech or large vocabularies requires advanced models like HMMs or deep neural networks.
- The lab reinforced understanding of both machine learning workflows and speech feature engineering, preparing for more advanced speech recognition tasks.